

You have $N \times C \times H \times W = 2 \times 2 \times 2 \times 2$ (That means batch size = 2, channel = 2, height = 2 and width = 2) Learned parameters: $\gamma = 2$, $\beta = 1$ for all positions. Use $\epsilon = 0$. Now, apply batch normalization for the following data:

	Channel 1		Channel 2	
Sample 1	1	1	3	3
	1	1	3	3
Sample 2	9	9	11	11
	9	9	11	11

Answer :

Channel_1 – values: {1, 1, 1, 1, 9, 9, 9, 9}

$$\mu_{C1} = (1 \times 4 + 9 \times 4) / 8 = 40 / 8 = 5$$

$$\sigma^2_{C1} = [(1-5)^2 \times 4 + (9-5)^2 \times 4] / 8 = (64 + 64) / 8 = 16 \rightarrow \sigma_{C1} = 4$$

Channel_2 – values: {3, 3, 3, 3, 11, 11, 11, 11}

$$\mu_{C2} = (3 \times 4 + 11 \times 4) / 8 = 56 / 8 = 7$$

$$\sigma^2_{C2} = [(3-7)^2 \times 4 + (11-7)^2 \times 4] / 8 = (64 + 64) / 8 = 16 \rightarrow \sigma_{C2} = 4$$

Block	x	$\hat{x} = (x - \mu) / \sigma$	$\hat{y} = 3\hat{x} + 2$
S1, C1	1	$(1-5)/4 = -1$	-1
S2, C1	9	$(9-5)/4 = 1$	5
S1, C2	3	$(3-7)/4 = -1$	-1
S2, C2	11	$(11-7)/4 = 1$	5

More to do: solve the same problem for $\epsilon = 0.5$